

AMAZON ONLINE PROFITABILITY ANALYSIS

Identifying Key Factors of Profitable and Non-Profitable Orders

ABSTRACT

This report analyzes Amazon online order data to find what factors are linked to profit and loss. The team uses Tableau visuals and regression analysis to find patterns by location, product type and more.

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Introduction

The data used in this report came from 9,890 Amazon online orders from 2016 to 2019. Amazon's leadership team can use this analysis to better understand which factors are associated with profitable and non-profitable orders and to make more informed pricing, promotion, and regional strategy decisions. This report analyzes profitability using Tableau data visualizations and multiple regression analysis. The results showed that revenue had a positive and statistically significant relationship with profit. Quantity and discount percentage had statistically significant negative relationships with profit. Based on these findings, the leadership team should review discount strategies, monitor quantity-based promotions, and pay closer attention to lower-profit regions and product categories. This report first presents the Tableau data exploration, then explains the regression analysis, discusses the relevance of the findings, and ends with recommendations and next steps.
[1]

Data Exploration

Click here for Average Profit Tableau visualization

This visualization includes a drop-down menu that can switch the categories the graph shows. The profit dashboard showed that profitability varied across the different product types. Copiers had the highest average profit per order, at approximately $816. Tables and bookcases had a negative average profit which shows some subcategories are more profitable than others. The regional graph showed that the West and East regions had the highest average profit and Central had the lowest. The time graph showed that average profit increased from 2016 to 2018 but then declined in 2019. This suggests profitability was on the rise for a couple of years and then dropped in the final year.

Click here for Revenue by State Tableau visualization

The revenue dashboard showed that average revenue varied by location. The city dot map showed larger revenue in many major cities like Vallejo and Waukesha. The state-filled map showed some states had a higher revenue than others with values ranging from about $17 to $724. These visuals help Amazon's leadership team compare where high value orders occur with where higher profitability is.

[1] ChatGPT was used to review this introduction for clarity, grammar, professionalism, and alignment with assignment criteria. Final edits were made by the project team.

Statistical Analysis

Data Overview

The dataset used in this analysis consisted of Amazon online orders from 2016-2019, with the final regression model using 9,890 observations. Six observations were removed from the Categorical Order Info data due to missing data values that we discovered during the data cleaning process. The raw data included variables of revenue, quantity purchased, discounts, shipping methods, customer segments, product categories, regions, cities, and states. Multiple dummy variables were created in Excel so that categorical data, like regions and shipping methods, could be included in the regression analysis. The final regression model used profit as the dependent variable and included revenue, quantity, discount percentage, regional dummy variables, and state dummy variables. The descriptive statistics and final regression model are shown in Figure 1 in the appendix.

Before conducting the statistical analysis, the dataset was reviewed for insignificant variables. The dummy variables representing weekend effects, holiday season effects, shipping methods, customer segments, subcategories, and city effects were tested in separate model specifications. However, these variables were statistically insignificant and did not meaningfully improve the explanatory power of the model. Therefore, they were excluded from the final specification.

Population Regression Equation

Estimated Regression Equation

R^2 Interpretation

The final model had a R^2 of 0.2724 which means the model explains about 27.24% of the variation in profitability. While this explains some of the variation, the quality of the model also depends on how well regression assumptions are satisfied. The residual plots and correlation matrix suggest that the assumptions were reasonably satisfied with no major violations.

Standard Error Interpretation

The final model had a standard error of 200.59. This means, on average, the model's profit predictions are off by about $200.59. This is useful for finding broader patterns but should not be treated as a perfect prediction.

Significance Testing at 10% Level

The analysis found that revenue, quantity, discount percentage, the Central regional dummy variable, and the California dummy variable were statistically significant because their p-values were below the 10% significance level. Other variables, such as West and East, were not statistically significant. However, they were still used in the model to show comparisons across regions.

Coefficient Interpretation

Profit increases by $0.18 for every $1 increase in revenue, on average and all else constant.

Profit decreases by $3.01 for every 1 unit increase in quantity purchased, on average and all else constant.

Profit decreases by $2.39 for every 1% increase in discount offered, on average and all else constant.

Profit is $6.66 higher for orders from the West region than orders from the baseline region, on average and all else constant.

Profit is $0.54 lower for orders from the East region than orders from the baseline region, on average and all else constant.

Profit is $17.97 higher for orders from the Central region than orders from the baseline region, on average and all else constant.

Profit is $13.04 lower for orders associated with California than orders from the baseline state, on average and all else constant.

Detection of Heteroskedasticity and Non-Linear Patterns

Residual plots were visually inspected to evaluate the assumptions of linear regression, including homoskedasticity and linearity. The residuals appeared to be generally randomly distributed around zero, and no strong nonlinear patterns were observed. Some variation in residual spread was visible, particularly for Revenue, suggesting the possible presence of mild heteroskedasticity. Several outliers were also observed, which is common in large business datasets.

See Figure 2 for Residual Plots

Detection of Endogeneity/Omitted Variables and Multicollinearity

The correlation matrix showed relatively low correlations among the main explanatory variables, indicating that multicollinearity is not a major concern in the model. The model included several profitability factors such as revenue, quantity, discounts, and more that reduces the chance that important profitability patterns were excluded.

See Figure 3 for Correlation Matrix

Relevance of Findings

From the results, we found several important trends for the Amazon leadership team. We found that revenue has a positive relationship with profit, meaning larger sales increase profitability. However, quantity and discounts had negative effects on profit. This suggests that selling more products does not always lead to higher profit. Offering larger quantity orders have heavier discounts, which as the discount percentage increases, profit decreases. These factors are important because it shows that discount and quantity strategies should be reviewed, as while discounts may increase quantity sold, they can reduce profitability.

One surprising finding we found was that quantity was negatively related to profit. Typically, a company may expect that selling more units would increase profit, and that the data should back that theory up. However, our models suggest that higher quantity orders may lower profit when other factors are controlled for.

Some variables were not significant and did not remain in our final model but should still be mentioned. For example, shipping method, customer segment, product category, weekend orders, and holiday season variables may still be useful for business decisions, even if they were not statistically significant in the final model.

Recommendation

Amazon's leadership team should use these findings to improve pricing, discounts, and regional strategy. Since discounts were negatively related to profit, Amazon should review how often discounts are used and whether certain promotions are reducing profit more than expected. Amazon could test some more targeted promotions that focus on higher margins or demanded products, rather than general sales throughout the year. Amazon should also investigate quantity-based offers, such as "buy one, get 50% off" promotions. This is because quantity was negatively related to profit, the leadership team should make sure that larger orders are not lowering margins.

The geographic results also suggest that Amazon should continue looking into profitability by region and state. If certain areas are more profitable than others, Amazon should see if those differences are caused by shipping distance, warehouse locations, or demand. This could help the leadership team make better decisions about regional distribution and potential promotions to boost revenue.

For future testing, Amazon should collect more data, which could include shipping costs, fulfillment costs, product margins, and advertising expenses. These variables could improve the predictive power of the model and help explain more of the variation in profit, both positive and negative. Future analysis could also test other variables, such as whether discounts affect profit differently by region, product category, or quantity.

Conclusion

The analysis showed important patterns that could help Amazon improve profitability. The regression model showed that higher revenue was linked to higher profits, while larger quantities and bigger discounts lowered profits. The results also showed differences across regions and states, which might have an impact on overall performance. Based on these findings, Amazon should take a closer look at its discount strategies, as well as see if large quantity orders are generating larger profits, if any at all. While the model explained a portion of the changes in profitability, having more variables in the future could show a better picture of the factors affecting profit. [2]

Should there be any questions, Amazon's leadership team may contact the team at:

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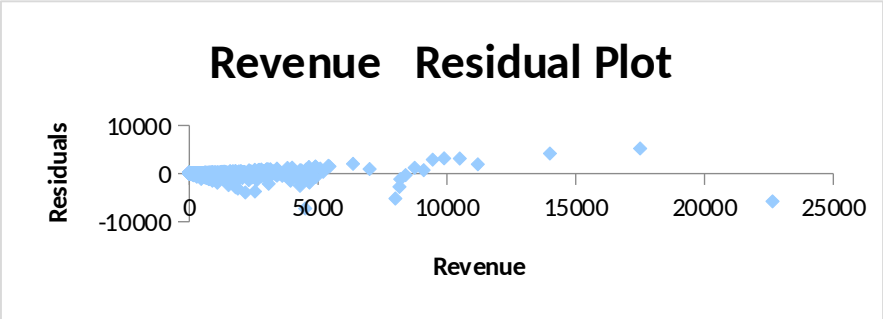
[2] ChatGPT was used to review this conclusion for clarity, grammar, professionalism, and alignment with assignment criteria. Final edits were made by the project team.

Appendix

Figure 1 – Descriptive Statistics and Final Regression Output

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Figure 2 – Residual Plots







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Figure 3 – Correlation Matrix

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Sources Consulted

OpenAI. (2026). *ChatGPT* (May 10 version) [Large language model].

https://chat.openai.com/chat